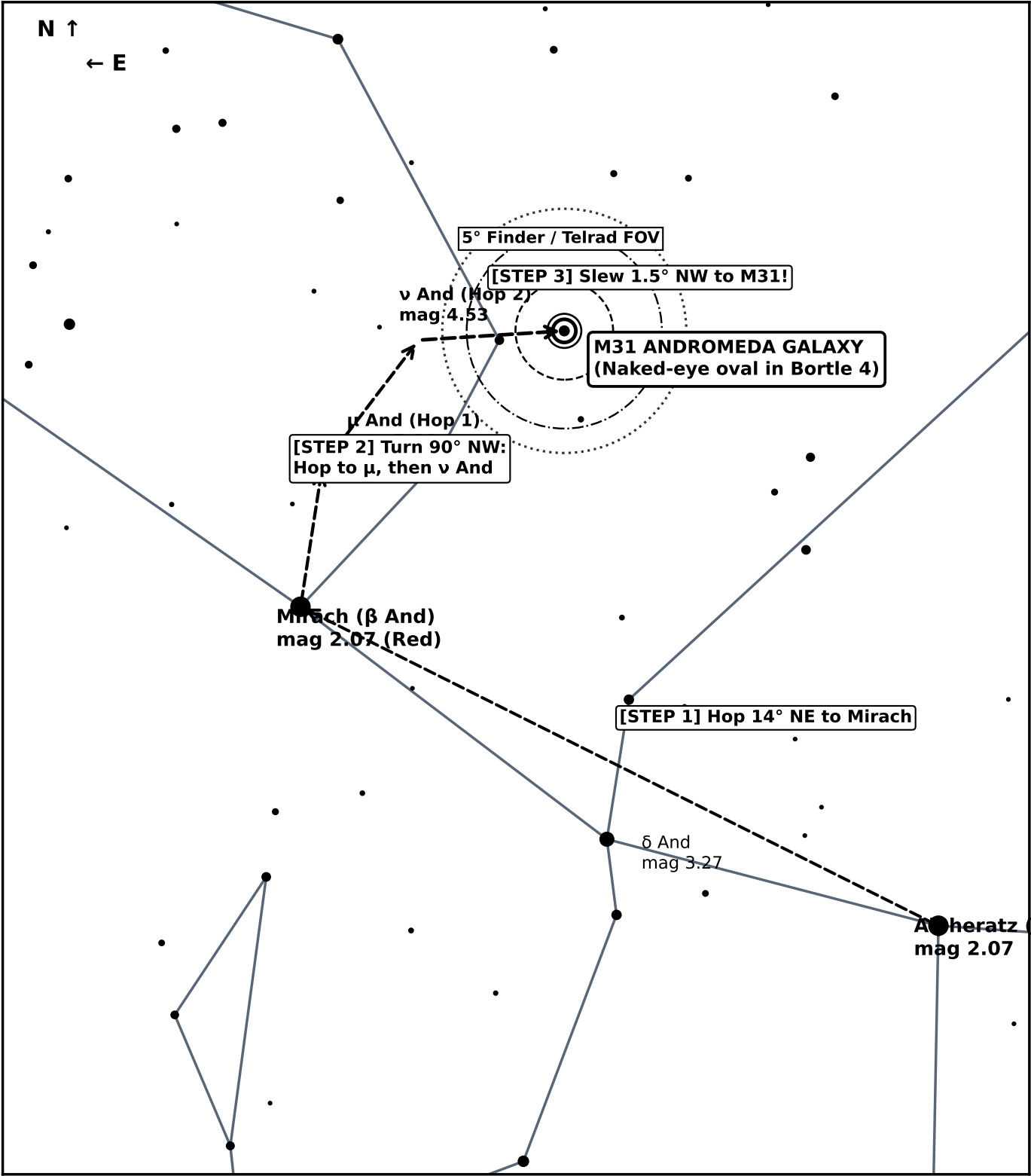


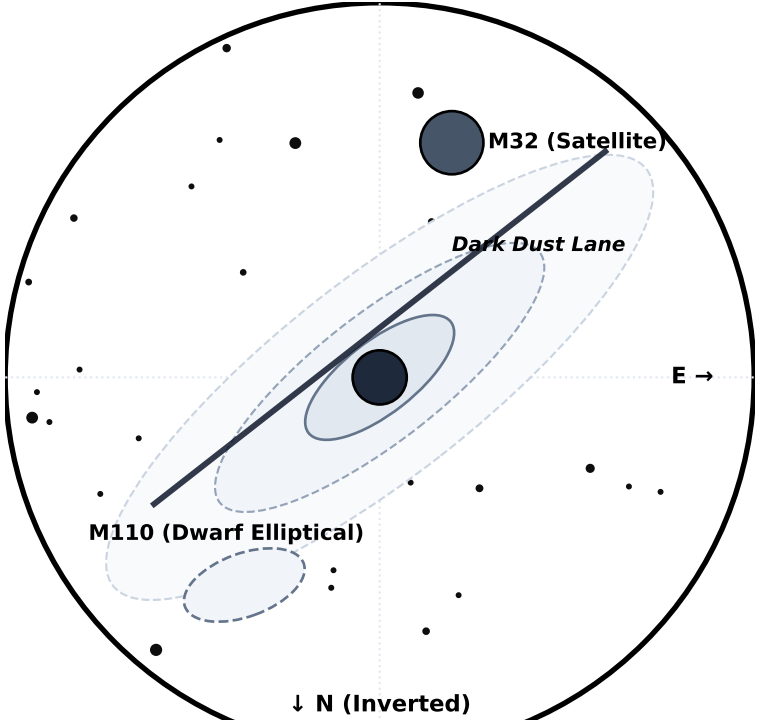
STAR-HOPPING FINDER CHART 4: M31 (ANDROMEDA GALAXY), M32 & M110 — ANDROMEDA

Location: Dvigrad (Bortle 4, SQM 21.0) | Sep 12, 2026 • 21:15 CEST | Sky-Watcher 200P Dobsonian (8" f/6, 1200mm FL) | B/W Toner-Saver Edition

VIEWPORT A: WIDE-FIELD STAR-HOPPING CHART (Upright Naked-Eye / Finder Orientation: N ↑, E ←)



VIEWPORT B: TELESCOPE EYEPIECE SIMULATION (Negative Inverted 180°: N ↓, E →)



20mm Eyepiece (60× Magnification, 50' True FOV)

VIEWPORT C: TARGET DOSSIER & STAR-HOPPING INSTRUCTIONS

OBJECT DOSSIER: M31 (NGC 224), M32 & M110

- Constellation: Andromeda | Type: Giant Spiral Galaxy
- Magnitude: 3.4V | Surface Brightness: 13.5 mag/arcmin²
- Apparent Size: 180' × 63' (~3°) | Distance: 2.54 million ly
- Tonight's Ephemeris at Dvigrad (Sep 12, 2026):
 - 20:00 CEST: Alt 24.7°, Az 52° (Rising in ENE)
 - 21:15 CEST: Alt 36.2°, Az 65° (Prime Contrast!)
 - 22:30 CEST: Alt 48.6°, Az 78° (Superb & High)

OBSERVING STRATEGY & EYEPIECES:

- 20mm Eyepiece (60×): Mandatory! With a 50' field, you can sweep across the immense galactic disk and examine the starlike active galactic nucleus and companion M32.
- Dark Dust Lane: Look just outside the central core along the northwestern edge; under Dvigrad's Bortle 4 sky, the silhouette dust absorption lane is unmistakable with averted vision.
- Companions: M32 is a bright, compact elliptical 24' south; M110 is a soft, diffuse oval cloud 35' northwest.

STEP-BY-STEP STAR HOP NARRATIVE:

- Naked-Eye: Find the Great Square of Pegasus. Identify the northeast corner star Alpheratz (α And).
- Follow the Andromeda chain: hop 7° to δ And, then 7° to reddish Mirach (β And, mag 2.1).
- Turn 90° NW: Hop 3.5° to μ And, then 3.5° to ν And.
- Slew 1.5° further NW: M31 fills the finder and eyepiece!